



Review

Scheduling jobs on identical parallel machines with unequal future ready time and sequence dependent setup: An experimental study

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ABSTRACT

This paper studies the performances of Apparent Tardiness Cost based (ATC-based) dispatching rules in minimizing the total weighted tardiness on identical parallel machines with unequal future ready time and sequence dependent setup.

In experiments, the sequence dependent setup is divided into two categories: the continuous sequence dependent setup ($s_{ij,con}$) and the separable sequence dependent setup ($s_{ij,sep}$). The former needs the presence of parts to setup on the machine, while the latter does not need it and allows machine to be idle between the setup and the processing of the part. Experiments show that Modified Apparent Tardiness Cost with Setup and Ready time (MATCSR) and Apparent Tardiness Cost with Separable Setup and Ready time (ATCSSR) outperform other existing ATC-based rules in reducing the total weighted tardiness for problems ($Pm|r_j, s_{ij,con}|\sum w_j T_j$ and $Pm|r_j, s_{ij,sep}|\sum w_j T_j$) where the sequence dependent setups are continuous and separable, respectively.

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1. Introduction

Scheduling jobs on parallel machines is one of the significant and natural extensions of the single machine scheduling, which received much attention in industry and academia. When production volume is large, management often uses multiple machines, which may be the same, similar, or totally different. Allahverdi and Mittenenthal (1996) group parallel machines into three categories: identical parallel machines, where the processing time of a job is the same on all machines, uniform parallel machines, where the processing time of a job is determined by the speed factor of the machine, and unrelated parallel machines, where the processing time of a job on machines can be different in an arbitrary way. The objectives of parallel machines scheduling are mostly either flow time-related or tardiness-related; the former tries to reduce the waiting time of parts or WIP, the latter tries to meet the due dates of orders. Wisner and Siferd (1995) say that meeting due dates is the most important scheduling objective, and only 58% of industrial planners manage to meet the due dates.

In literature, most research assumes setup time can be either ignored or included in a part's processing time. This assumption is reasonable only when the setup time is independent from the job sequence. Sequence dependent setup, in which the length of setup time of a job depends on its immediately preceding job, is common and often important in production. Examples are found in petroleum production plants, printing plants, car spraying facilities, metallurgical industries and textile dying plants (Luo et al. 2006, Arroyo et al., 2009). Based on the report of Panwalkar et al. (1973), about 70% of schedulers claim that a quarter of jobs they schedule cannot ignore setup time. Krajewski et al. (1987) claim that effective management of sequence dependent setup is one of the critical factors to improve the performances of a manufacturing system.

The scheduling problem of parallel machines is important also in a service system such as a hospital or a car repair shop, as well as a manufacturing system. The setup for surgical operations often does not need patients at the sites and the setup can be sequence dependent. Similarly, in a car repair shop, the setup also does not need cars in bays and setup also can be sequence dependent (Kim et al., 2003b). Although the result of this paper applies to service industry, we use the terminology of manufacturing without confusion.

Yue and Jang (2011) first classify the sequence dependent setup into two categories based on whether or not the setup of the next selected future job is allowed to start when a machine becomes idle, even before the next job is ready to be processed. If the setup can start before the part becomes ready, it is Separable Sequence Dependent Setup (SSDS); otherwise it is Continuous Sequence Dependent Setup (CSDS). An example of the SSDS is color changing in a painting or a product coloring shop, where the color changes from a dark color to a light color takes longer time than in the opposite case. Often setup for a new color does not require parts on a machine. Press die change and machine tooling change are additional examples at the machine level, and assembly line setup change is an example at the production line level. These setups do not need parts to be ready, and the length of setup times depends on the last scheduled part.

This paper studies the performances of Apparent Tardiness Cost based (ATC-based) dispatching rules on identical parallel machines. It is organized as follows: in Section 2, we describe the problems. Studies of parallel machines scheduling with sequence dependent

setup and ATC-based rules are reviewed in Section 3. Section 4 discusses the grid approach of ATC-based rules and experiment designs. Sections 5 and 6 evaluate the performances of ATC-based rules under the conditions where the setups are CSDS and SSDS, respectively. Section 7 records the computation time of scheduling. Finally, conclusions and future research are discussed in Section 8.

2. Problem description

This paper considers two problems: $Pm|r_j, s_{ij}, con|\sum w_j T_j$ and $Pm|r_j, s_{ij}, sep|\sum w_j T_j$. The first problem is stated as: there are n jobs arriving to m identical parallel machines at different times. Each job j has its ready time (r_j), processing time (p_j), due date (d_j), and job weight (w_j). The setup time (s_{ij}) of each pair of jobs i and j is sequence dependent and the continuous type, *con*. In general, s_{ij} is not equal to s_{ji} . The objective is to minimize the total weighted tardiness of jobs, $\sum_{j=1}^n w_j T_j$, where T_j is the tardiness of job j , $\max\{0, C_j - d_j\}$, and C_j is the completion time of job j . The second problem is the same as the first problem except that the setup is changed into the separable type, *sep*. Both problems assume the following:

- The job attributes (p_j , d_j , w_j , r_j , s_{ij}) are known in advance.
- The machines are parallel identical.
- Each machine can process at most one job at each time.
- Job preemption is not allowed.
- Production interruptions such as machine breakdown and order cancellation do not happen.

3. Literature review

This paper focuses on parallel machines scheduling with sequence dependent setup, related works are reviewed by the objective criteria.

3.1. Completion-time-related-criteria

Kurz and Arskin (2001) present an integer programming and a heuristic procedure to minimize the sum of completion times of all jobs on identical parallel machines. Their heuristic first assigns jobs to the machine, and then determines the optimal job sequence of each machine by solving the traveling salesman problem. Weng et al. (2001) minimize the total weighted completion time on unrelated parallel machines. Several heuristics are tested. The best heuristic assigns jobs based on the smallest ratio of a job's processing time plus setup time to its weight. Fowler et al. (2003) use a hybrid genetic algorithm to minimize total weighted completion time. Their genetic algorithm first assigns jobs to machines, and a few different dispatching rules schedule assigned jobs on each machine. Chen et al. (2003) propose a branch and bound algorithm to minimize total weighted completion time on identical parallel batch machines. A computational analysis shows that it can optimally solve medium-sized problems within reasonable computation time. Munoz et al. (2005) propose a constructive heuristic to assign jobs iteratively with the minimum adjusted processing time (sum of setup time and processing time). Their approach assigns jobs either at the beginning or end of the created partial schedule. Anglian et al.

(2005) use fuzzy mathematical programming to minimize total setup time on identical parallel machines.

3.2. Due-date-related criteria

ATC-based rules are known for making good schedules in minimizing tardiness under diverse conditions of the scheduling problems, such as single machine scheduling problems (Rachamadugu and Morton, 1982; Morton and Pentico, 1993; Lee et al., 1997; Yue and Jang, 2011) and job shop scheduling problems (Vepsalainen and Morton, 1987; Mason et al., 2002).

For parallel machines scheduling, Lee and Pinedo (1997) propose Apparent Tardiness Cost with Setups (ATCS) for identical parallel machines. They first use four factors to characterize a problem instance, and then calculate the values of the scaling parameters in the index of ATCS. Park et al. (2000) extends the work (Lee and Pinedo, 1997) and introduce the setup time range factor. Five factors are used as inputs by the neural network to decide the values of the scaling parameters. Lars et al. (2006) propose a heuristic to minimize the total weighted tardiness on identical parallel batch machines. Their ATC-based heuristic considers incompatible families and unequal ready time. However, their heuristic does not consider the sequence dependent setup. Apparent Tardiness Cost with Setups and Ready times (ATCSR, Pfund et al., 2008) is the latest ATC-based dispatching rule to minimize the total weighted tardiness on identical parallel machines with unequal ready time. It uses three scaling parameters, k_1 , k_2 and k_3 and search solution (schedule) in a 3d space.

Besides the above-mentioned dispatching rules, meta-heuristics are effective methods to find near optimal solutions for the due-date-related problems. However, they tend to induce big computational burdens and are suitable only when no quick solutions are needed. Tarnimi and Rajan (1997) propose a genetic algorithm for uniform parallel machines scheduling. They dynamically modify mutation, crossover, and insertion rates. Kim et al. (2002) consider unrelated parallel batch machines scheduling where jobs within a family have the same due date. Eom et al. (2002) consider unrelated parallel batch machines scheduling, and propose a three-stage method in which the last stage is tabu search. For the same problem, Kim et al. (2003a) test four heuristics: the earliest weighted due date, the shortest weighted processing time, the two-level batch scheduling heuristic, and the simulated annealing method. Their test shows that simulated annealing outperforms the other heuristics. Fowler et al. (2003) use a hybrid genetic algorithm to minimize total weighted tardiness on identical parallel machines. Chen (2009) proposes a hybrid method for unrelated parallel machines. They use ATCS to generate an initial solution and then use simulated annealing to improve the results. Their experiments show that simulated

annealing effectively improves the initial solution obtained by ATCS.

Some studies consider two due-date-related objectives simultaneously. These studies are relevant to Just In Time (JIT) production. Heady and Zhu (1998) minimize the sum of earliness and tardiness on identical parallel machines. They compare the result of the proposed heuristic with the optimal solution for a small size problem. Radhakrishnan and Ventura (2000), Kesen and Baykoc (2007) minimize the sum of weighted earliness and weighted tardiness on uniform parallel machines. Two mathematical programming models are proposed to find optimal solution. For other objectives, Hirashi et al. (2002) address identical parallel machines scheduling in maximizing the weighted number of non-tardy jobs.

Except for ATCSSR (Yue and Jang, 2011), all heuristics surveyed in this section consider CSDS only. Table 1 summarizes the ATC-based dispatching rules in literature.

More literature on general scheduling problems with setups are surveyed by Allahverdi et al. (1999, 2008). In the latter work, three criteria are used to group the selected papers: (1) batching and non-batching production, (2) sequence independent setup and sequence dependent setup, and (3) shop environments, such as job shop and flow shop.

4. The grid approach and design of experiment

This section first discusses the grid approach used for ATC-based rules and then introduces the experiment designs of the performance tests.

4.1. The grid approach and distribution of best grid points

The grid approach attempts to find a good schedule by trying multiple combinations of parameter values (e.g., the k values of the index equations of ATC-based rules), and is used for parallel machines scheduling (Park et al., 2000; Pfund et al., 2008; Rene and Lars 2009, 2011) as well as for single machine problems. Like in all other search procedures, the search space is important for the grid method; this section tests the distributions of the best values of the scaling parameters, k_1 , k_2 , and k_3 of ATCSR by the grid method.

To find the distribution of good grids (or combination of k values) of the parallel scheduling problems, we use the grid points by Yue and Jang (2011). They use the same number of grid points as that of Pfund et al. (2008) for the k values (22 values of k_1 , 11 values of k_2 and 13 values of k_3), but covers larger range for each scaling parameter (grid size is increased). In this test, ATCSR checks 3146 ($22 \times 11 \times 13$) grids and records the best combination of k values out of 3146 grids for each of the 5103 problems, which is generated in Section 4.2. The test result is given in

Table 1
ATC-based dispatching rules.

Authors	Rule name (objective function)	Number of parameters	Future ready time	Sequence dependent setup	Environment
Rachamadugu and Morton (1982)	ATC (TWT)	1 (k)	No	No	Single machine
Lee et al. (1997); Lee and Pinedo (1997)	ATCS (TWT)	2 (k_1 and k_2)	No	No	Single machine/Parallel machines
Vepsalainen and Morton (1987)	MATC (TWT)	2 (b and k)	No	No	Flow shop, Job shop
Morton and Pentico (1993)	X-RM (TWT)	2 (B and k)	Yes	Yes	Single machine
Mason et al. (2002)	BATCS (TWT)	2 (k_1 and k_2)	Yes	Yes	Batch machine
Lars et al., 2006	BATC-based heuristic (TWT)	1 (k_1)	Yes	Yes	Parallel batch machines
Pfund et al. (2008)	X-RMmod (TWT)	3 (B , k_1 and k_2)	Yes	Yes	Parallel machines
Pfund et al. (2008)	BATCSmod (TWT)	2 (k_1 and k_2)	Yes	Yes	Batch machine
Pfund et al. (2008)	ATCSR (TWT)	3 (k_1 , k_2 and k_3)	Yes	Yes	Parallel machines
Yue and Jang (2011)	MATCSR (TWT), ATCSSR (TWT)	3 (k_1 , k_2 and k_3)	Yes	Yes	Single machine

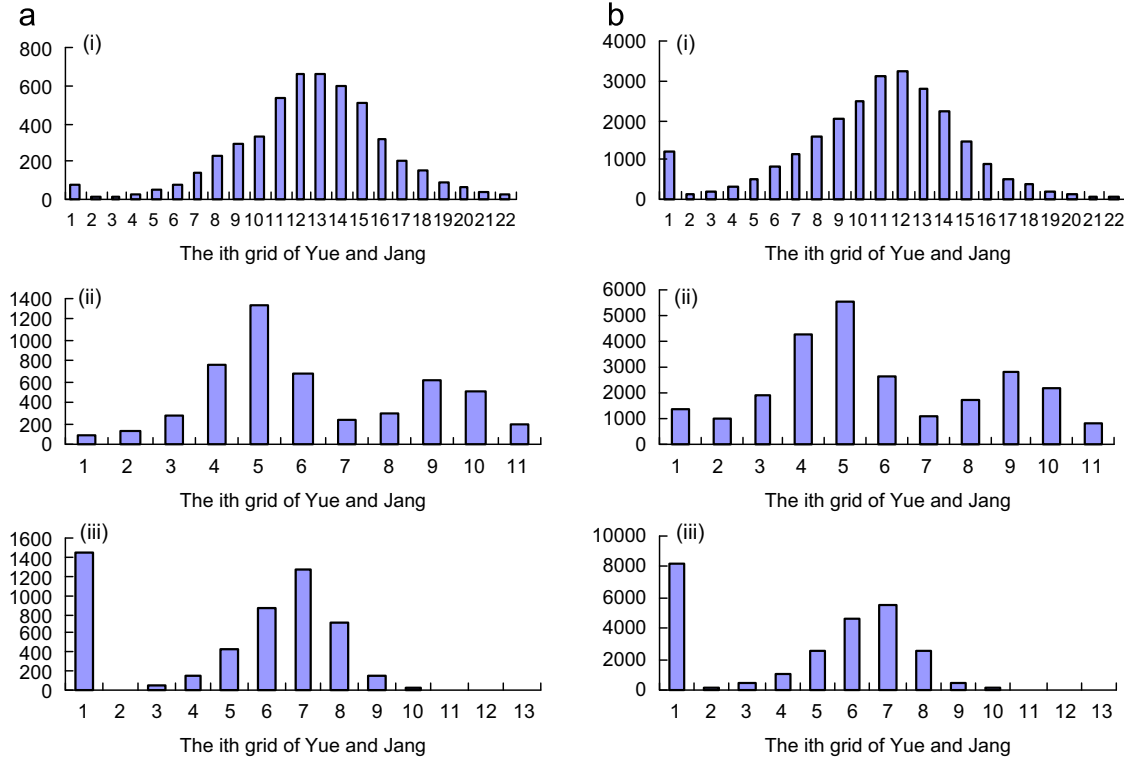


Fig. 1. Distribution of the best k 's of ATCSR (y axis: frequency). (a) distribution of the best k 's: (i) distribution of the best k_1 Pfund's range (6th–16th), (ii) distribution of the best k_2 Pfund's range (3th–8th), (iii) distribution of the best k_3 Pfund's range (2th–8th) and (b) distribution of the best five k 's: (i) distribution of the best five k_1 's Pfund's range (6th–16th), (ii) distribution of the best five k_2 's Pfund's range (3th–8th), (iii) distribution of the best five k_3 's Pfund's range (2th–8th).

Fig. 1(a), where the x axis is the grid number of each scaling parameter, and y axis is the frequency of the best scaling parameter. In the figure, we observe that there are about 85%, 70%, and 68% of the best k_1 , k_2 , and k_3 values located in the range of Pfund. We also record the distribution of the best five combinations of scaling parameters for each of 5103 problems. This analysis is helpful because the grid method often does not give the best k values in the search range due to the positive value of the grid size. Fig. 1(b) shows there are about 85%, 68%, and 66% of the best five k_1 , k_2 , and k_3 values located in the range of Pfund. These tests show that the coverage of Pfund's current grid ranges is somewhat "limited" but acceptable.

To comment on their settings, Pfund et al. (2008) claim that "The grid used for comparable algorithms such as ATCS (Lee and Pinedo, 1997) was studied and additional values were added to ensure that the frequently occurring values were not the extreme values in the grid." In regards to settings of Pfund, Rene and Lars (2009) comment that the settings of Pfund are large and time consuming. They suggest using smaller grid ranges (a subset ranges from that of Pfund) with less number of grid points (seven values for k_1 , four values for k_2 , and five values for k_3). Rene and Lars (2011) use settings of Pfund to generate initial solution for their new heuristic. Because ATCSR is the latest ATC-based index proposed in the literature for identical parallel machines scheduling and ATCSR uses Pfund's grid setting, we use that setting instead of the setting of Yue and Jang (2011) to compare the performances under a more objective performance test setting that is proposed not by the authors.

Set k_1 :

{0.2,0.6,0.8,1,1.2,1.4,1.6,1.8,2.2,4.2,8.3,2.3,6.4,4.0,4.4,4.8,5.2,5.6,6.4,6.8,7.2}

Set k_2 :

{0.1,0.3,0.5,0.7,0.9,1.1,1.3,1.5,1.7,1.9,2.1}

Set k_3 :

{0.001,0.0025,0.004,0.005,0.025,0.04,0.05,0.25,0.25,0.4,0.6,0.8,1,1.2}

4.2. Benchmark methods and design of experiment

Five ATC-based dispatching rules, BATCS, BATCSmod, ATCSR, MATCSR and ATCSSR are studied in the experiments. When BATCS and BATCSmod are used as benchmark methods, their batch size is set at 1. ATCSR, MATCSR and ATCSSR have three k values, while BATCS and BATCSmod have only two k values. These ATC-based indexes are given as below:

BATCS (Mason et al. 2002)

$$I_{\text{BATCS}}(t, i, j) = \frac{w_j}{p_j} \exp\left(-\frac{\max(d_j - p_j - r_j - t, 0)}{k_1 \bar{p}}\right) \exp\left(-\frac{s_{ij}}{k_2 \bar{s}}\right) \quad (1)$$

BATCSmod (Pfund et al., 2008)

$$I_{\text{BATCSmod}}(t, i, j) = \frac{w_j}{p_j} \exp\left(-\frac{\max(d_j - p_j + \max(r_j - t, 0), 0)}{k_1 \bar{p}}\right) \exp\left(-\frac{s_{ij}}{k_2 \bar{s}}\right) \quad (2)$$

ATCSR (Pfund et al., 2008)

$$I_{\text{ATCSR}}(t, i, j) = \frac{w_j}{p_j} \exp\left(-\frac{\max(d_j - p_j - \max(r_j, t), 0)}{k_1 \bar{p}}\right) \exp\left(-\frac{s_{ij}}{k_2 \bar{s}}\right) \times \exp\left(-\frac{\max(r_j - t, 0)}{k_3 \bar{p}}\right) \quad (3)$$

MATCSR (Yue and Jang, 2011)

$$I_{\text{MATCSR}}(t, i, j) = \frac{w_j}{p_j + s_{ij} + \max(r_j - t, 0)} \exp\left(-\frac{\max(d_j - p_j - s_{ij} - \max(r_j, t), 0)}{k_1 \bar{p}}\right) \times \exp\left(-\frac{s_{ij}}{k_2 \bar{s}}\right) \exp\left(-\frac{\max(r_j - t, 0)}{k_3 \bar{p}}\right) \quad (4)$$

Table 2

Factors and levels of the performance test (Pfund et al., 2008).

Factor name	Equation or meaning	Low level	Center level	High level
Job machine factor: μ	$\mu = n/m$	11	19	27
Setup severity factor: η	$\eta = \bar{s}/\bar{p}$	0.02	1.01	2
Due date tightness factor: τ	$\tau = 1 - \bar{d}/C_{\max}$	0.3	0.6	0.9
Due date range factor: R	$R = (d_{\max} - d_{\min})/C_{\max}$	0.25	0.63	1
Job availability factor: J_a	J_a = Number of available jobs at 0/n	0.2	0.5	0.8
Ready time factor: $r_{-}\tau$	$r_{-}\tau$ is a measure of how closely the ready times are placed to the due dates ^a	1	5.5	10

^a The ready time r_j is generated with uniform probability in the range $[d_j - r_{-}\tau^*p_j, d_j]$. If $(d_j - r_{-}\tau^*p_j)$ is less than 0, a range of $[0, d_j]$ is used for ready time generation.

Table 3

Number of jobs of the experiment.

Job machine factor: μ	Number of machines		
	Two machines	Four machines	Six machines
$\mu = 10$	20 jobs	40 jobs	60 jobs
$\mu = 20$	40 jobs	80 jobs	120 jobs
$\mu = 30$	60 jobs	120 jobs	180 jobs

ATCSSR (Yue and Jang, 2011)

$$I_{ATCSSR}(t, i, j) = \frac{w_j}{p_j + \max(s_{ij}, r_j - t)} \exp\left(-\frac{\max(d_j - p_j - \max(r_j, t + s_{ij}), 0)}{k_1 \bar{p}}\right) \times \exp\left(-\frac{s_{ij}}{k_2 \bar{s}}\right) \exp\left(-\frac{\max(r_j - t - s_{ij}, 0)}{k_3 \bar{p}}\right) \quad (5)$$

In above formulas, t is the decision time. k_1 , k_2 , and k_3 are scaling parameters. $\bar{p}$ and $\bar{s}$ are average of processing time and average of setup time, respectively. Specifically, the sequence dependent setup in formula (1)–(4) is the continuous type, while in formula (5), it is the separable type.

Three experiments are performed to evaluate the performances of ATC-based rules on identical parallel machines scheduling. The first experiment completely repeats Pfund's 3⁶ experimental design but consider more problems: Each of the six factors has three levels (low, center, and high). The number of machines is set at 5 ($m=5$). This experiment has 729 (3⁶) scenarios. In each scenario, seven cases, or problems, are randomly generated. In total, 5103 (729 $\times$ 7) problems are tested in the first experiment. Table 2 shows factors and levels of Pfund. We evaluate the performance of ATC-based rules by two types of tests, *the best of the best test* and *the territory test*, which are explained in detail in Sections 5.1.1 and 5.1.2.

In the second experiment, we focus on the effect of the number of machines. This is a 3⁷ experimental design extended from Pfund's 3⁶ experiment. A new factor called machine number factor is introduced. The low, center, and high levels of this factor are set at 2, 4, and 6, respectively. In addition, the low, center, and high levels of the job machine factor, μ , are set at 10, 20, and 30. The settings of the rest of the five factors (η , τ , R , J_a , and $r_{-}\tau$) are exactly the same as those of Pfund. Table 3 contains nine sections, and shows the total number of jobs considered in each section. There are 2187 (3⁷) scenarios in this experiment. Each scenario contains seven problems. In total, 15,309 (2187 $\times$ 7) problems are considered in the second experiment.

The WSPT terms of MATCSR and ATCSSR have new parts, $\max(r_j - t, 0)$ and $\max(s_{ij}, r_j - t)$, respectively, compared to the WSPT terms of BATCS, BATCSRmod, and ATCSR. The third experiment is relevant to the significance of these new parts. When the setups are continuous type, we compare MATCSR with and without the new part, $s_{ij} + \max(r_j - t, 0)$, with ATCSR for the same 5103 problems generated in the first experiment (3⁶ experiment of Pfund). When the setups are separable, ATCSSR with and without the new part, $\max(s_{ij}, r_j - t)$, is compared with ATCSR for the same 5103 problems.

5. Performance evaluation of ATC-based rules under continuous sequence dependent setup (CSDS)

This section shows results of the three experiments mentioned in Section 4 for the continuous sequence dependent setup cases.

5.1. Performance comparison of different ATC-based rules with CSDS

5.1.1. The best of the best test

For each scheduling problem, MATCSR and ATCSR make 3146 ($22 \times 11 \times 13$) grids, and generate the same number of schedules. For BATCS and BATCSRmod, 242 (22×11) grids and the same number of schedules are made. The schedule with the smallest total weighted tardiness among the 3146, or 242 schedules is selected as the corresponding rule's final schedule for the considered problem. We compare the best selected schedules, one from each scheduling method, and check which method gives the best schedule. We call this test *the best of the best test*.

In this test, MATCSR, ATCSR, BATCS and BATCSRmod have 3407 (66.76%), 2157 (42.27%), 88 (1.7%), and 30 (0.59%) times to get the best solution out of the 5103 problems generated in Section 4.2. The sum of percentage is more than 100% due to ties. This test shows that MATCSR yields more number of the best cases than ATCSR, which significantly outperforms BATCS and BATCSRmod.

Next, we compare the two best methods from the above experiment, ATCSR and MATCSR, and study the effect of each factor. Fig. 2 shows MATCSR outperforms ATCSR at all levels on all factors except on factor 1, setup severity. Especially, in the 1701 problems (1/3 of 5103) where the setup severity factor is low (average setup time is short, i.e., 2% of the average processing time), ATCSR slightly outperforms MATCSR (39% to 32% of 1701 cases); they show the same tardiness level for the rest cases (29% of 1701 cases). Amongst the total of 5107 problems, there are 2908 (57%) problems where MATCSR is better, 536 (11%) problems are tied, and 1659 (32%) problem where MATCSR is worse. For problems where MATCSR gives better results, the average of improvements over ATCSR is about 3%. For problems where ATCSR gives better results, MATCSR yields worse results at an average regression of about 2.1% over ATCSR.

5.1.2. The territory test

This section performs *the territory test* as follows: For each of the 5103 problems generated in 4.2, the results from MATCSR and ATCSR are compared at each of all 3146 grid points and the percentages of grid points at which MATCSR performs better than, equal to, and worse than ATCSR are recorded. The test uses the average values of the percentages of the seven problems of each scenario. This analysis is important in relation to the regression method because the k values estimated by the regression method are somewhat away from the best grid point in a random fashion. When the best k values are not estimated accurately, the scheduling method with a larger favorable territory is likely to give a better schedule. (On the other hand, when the regression

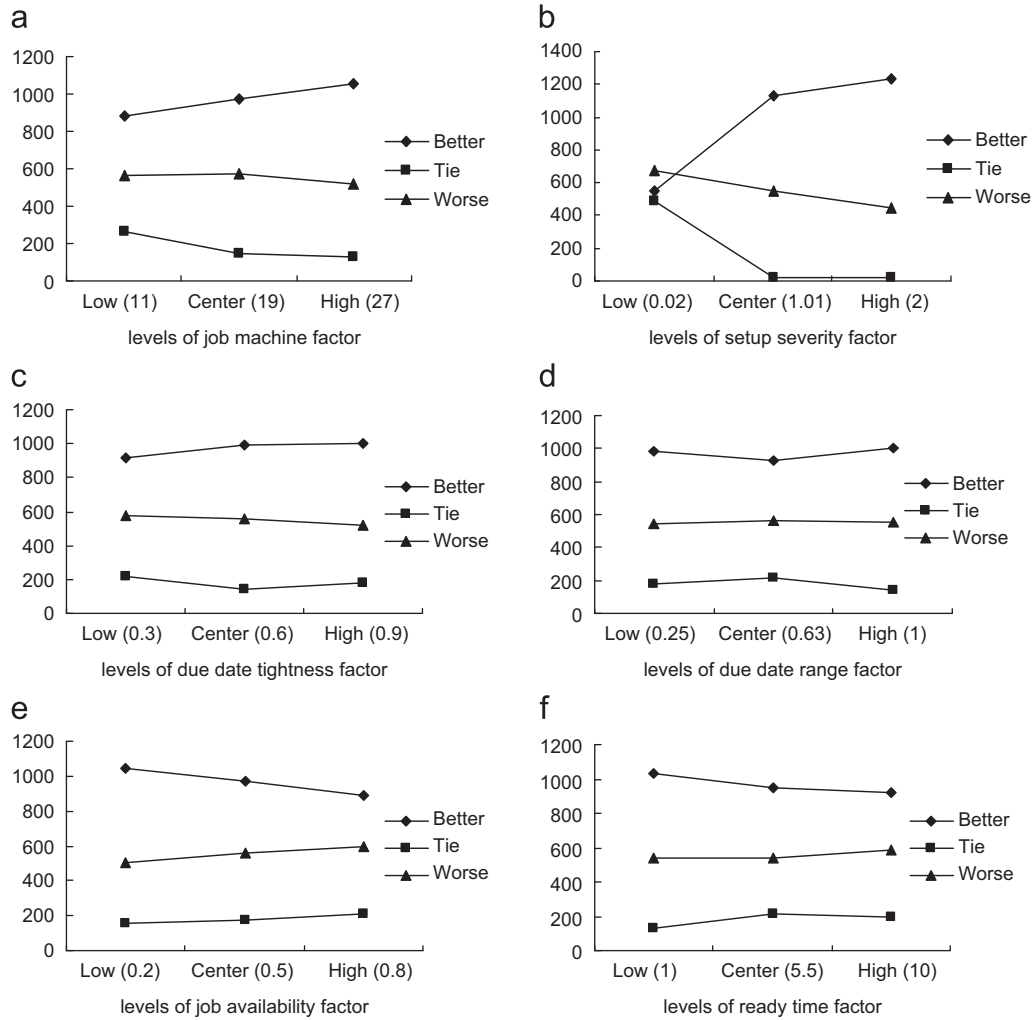


Fig. 2. Performance comparison of MATCSR against ATCSR (y axis: number of cases). (a) Effect of job machine factor, (b) Effect of setup severity factor, (c) Effect of due date tightness factor, (d) Effect of due date range factor, (e) Effect of job availability factor and (f) Effect of ready time factor.

equation gives the best k values accurately, the best of the best test introduced in the previous section is more relevant because it compares the best points selected by ATC-based rules.)

Fig. 3(a-i), (a-ii), and (a-iii) shows that when setup severity is at the high or center levels (scenarios 1 to 81 or scenarios 82 to 162, respectively) with 55, 95, and 135 jobs on 5 machines, respectively, MATCSR performs significantly better than ATCSR. It also shows that the number of jobs does not affect the test result much. When setup severity level is low (scenarios 163 to 243), many grid points yield the same total weighted tardiness for these two methods. The percentage of ties decreases when the number of jobs increases. It can also be noticed that when the setup severity level is low and due date range tightness is central or low (scenarios 190 to 243), MATCSR outperforms ATCSR again. Only when the setup severity level is low and due date tightness level is high, ATCSR is found to have less chance to outperform MATCSR in the territory test (scenarios 163 to 189).

5.2. Effect of the number of machines

Fig. 4(a) shows the effect of the number of parallel machines. It shows that when the level of the machine number factor, μ , increases, MATCSR outperforms ATCSR more. Additionally, for a given value of job machine factor, μ , using more machines yields more number of better cases.

5.3. Effect of the new part in the WSPT term of MATCSR, $s_{ij} + \max(r_j - t, 0)$

The result of the best of the best test summarized in Fig. 5(a) shows the importance of the new part in the WSPT term of MATCSR. Without $s_{ij} + \max(r_j - t, 0)$, the number of better cases is reduced and the number of tied cases is increased significantly.

6. Performance evaluation of ATC-based rules under separable sequence dependent setup (SSDS)

This section shows the results of the three experiments explained in Section 4 for the separable sequence dependent setup cases. Due to the lack of benchmark methods for the separable sequence dependent setup case, BATCS, BATCSmod, and ATCSR are slightly modified as follows and compared with ATCSSR: If a future job is selected to process next, its setup is allowed to start as soon as the last scheduled job is finished.

6.1. Performance comparison of different ATC-based rules with SSDS

6.1.1. The best of the best test

In this test, ATCSSR, ATCSR, BATCS, and BATCSmod have 3517 (68.92%), 1874 (36.72%), 101 (1.98%), and 45 times (0.88%) to get the best solution for the 5103 problems generated in Section 4.2.

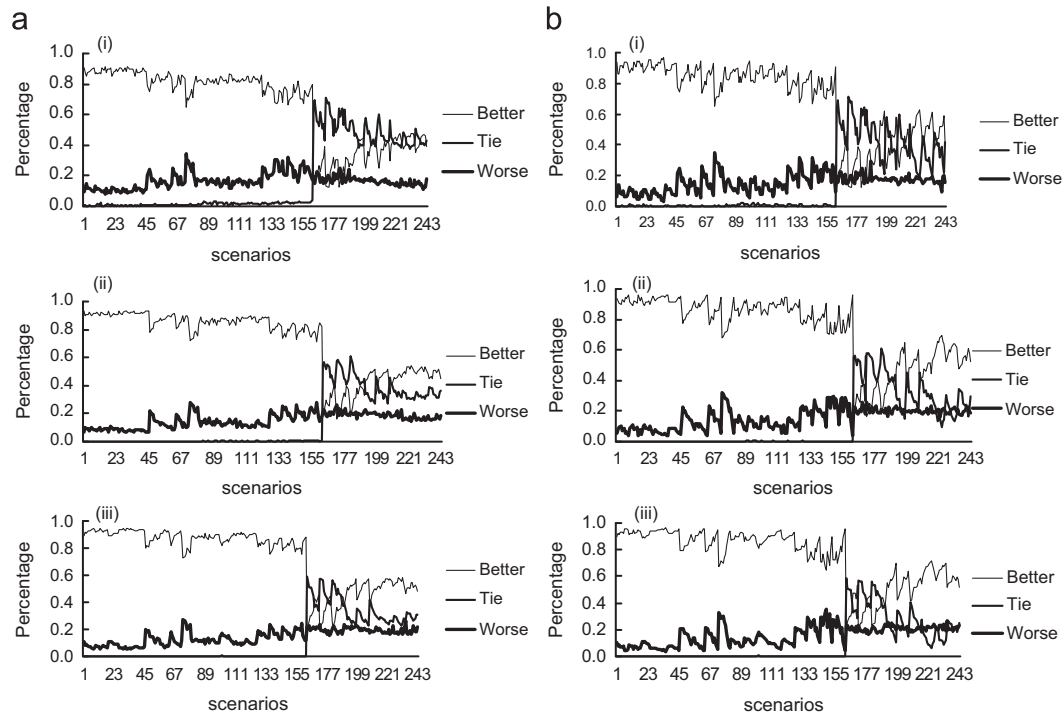


Fig. 3. Territory test. (a) MATCSR vs. ATCSR; (i) 55 jobs on 5 machines (ii) 95 jobs on 5 machines (iii) 135 jobs on 5 machines and (b) ATCSSR vs. ATCSR; . (i) 55 jobs on 5 machines (ii) 95 jobs on 5 machines (iii) 135 jobs on 5 machines.

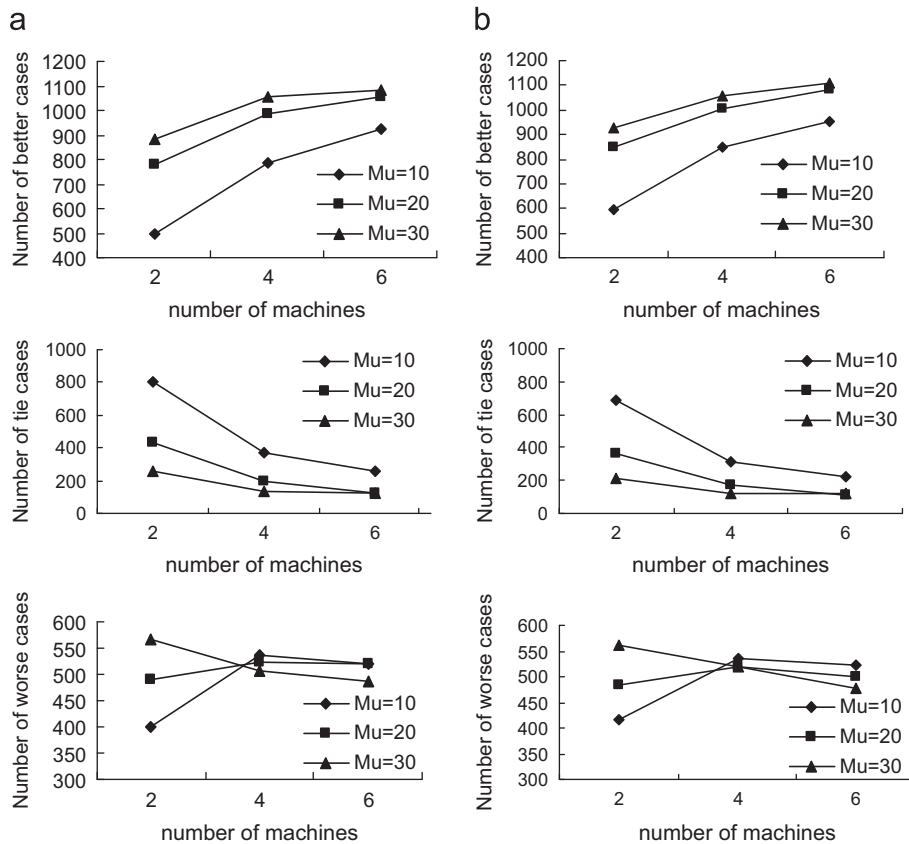


Fig. 4. The effect of the number of machines. (a) MATCSR vs. ATCSR and (b) ATCSSR vs. ATCSR.

The sum of percentage is more than 100% because of ties. This test shows that ATCSSR yields more number of the best cases than ATCSR, which significantly outperforms BATCS and BATCSmod.

Next, we compare the two best methods, ATCSR and ATCSSR, in more detail. Fig. 6 shows ATCSSR outperforms ATCSR at all levels on all factors, even when the setup severity is low (average

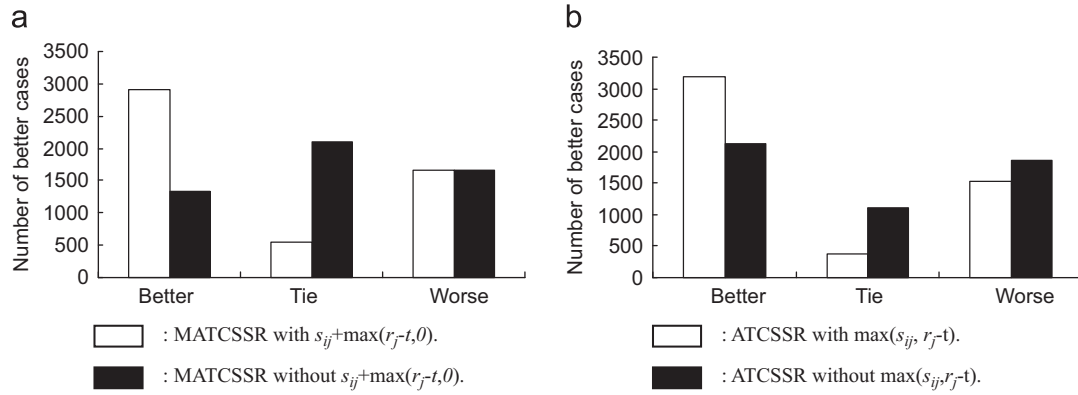


Fig. 5. Effect of new part in the WSPT term of MATCSSR and ATCSSR.

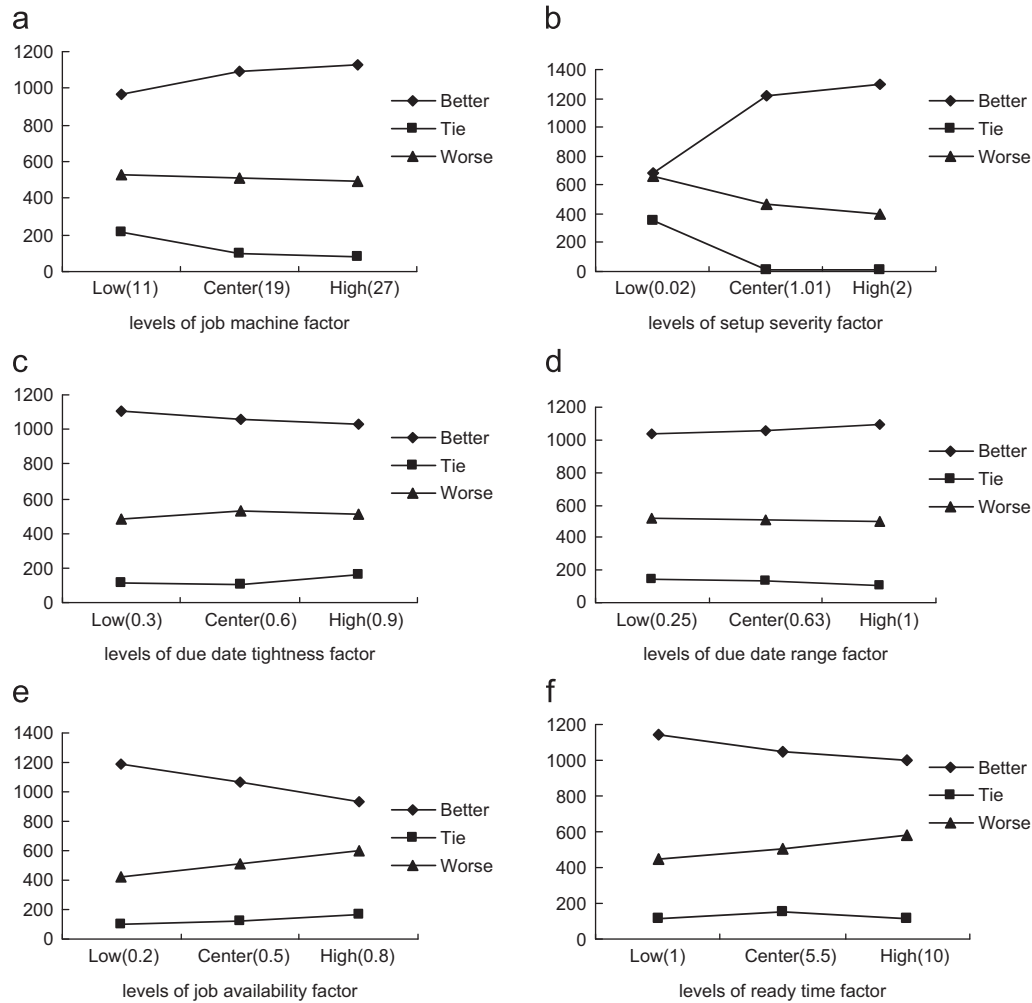


Fig. 6. Performance comparison of ATCSSR against ATCSR (y axis: number of cases). (a) Effect of job machine factor, (b) Effect of setup severity factor, (c) Effect of due date tightness factor, (d) Effect of due date range factor, (e) Effect of job availability factor and (f) Effect of ready time factor.

setup time is short, i.e., 2% of the average processing time). Amongst the total of 5107 problems, there are 3189 (62%) problems where ATCSSR is better, 385 (8%) problems are tied, and 1529 (30%) problem where ATCSSR is worse. For problems where ATCSSR gives better results, the average of improvements over ATCSR is about 3.2%. For problems where ATCSR gives better results, ATCSSR yields worse results at an average regression of about 2.0% over ATCSR.

6.1.2. The territory test

Fig. 3(b-i), (b-ii), and (b-iii) shows, when setup severity is at the high or center levels (scenarios 1 to 81 or scenarios 82 to 162, respectively) for 55 jobs, 95 jobs, and 135 jobs on 5 machines, ATCSSR performs significantly better than ATCSR, and the number of jobs does not affect the result significantly. When setup severity level is low (scenarios 163 to 243), many grid points yield the same total weighted tardiness. It can be noticed that

Table 4
The computation time.

Five machines	Methods with two scaling parameters (BATCS and BATCSmod)	Methods with three scaling parameters (ATCSR, ATCSR, and MATCSR)
55 jobs	Less than 1 s/problem	Less than 1 s/problem
95 jobs	Less than 1 s/problem	About 1.5 s/problem
135 jobs	Less than 1 s/problem	About 4.5 s/problem

when the setup severity level is low, and due date tightness level is central or low (scenarios 190 to 243), ATCSSR outperforms ATCSR again. Also, under this condition, the rate of ties decreases with increasing number of jobs. Only when the setup severity factor level is low and due date tightness level is high, ATCSR has less chance to outperform ATCSSR in the territory test (scenarios 163 to 189).

6.2. Effect of the number of machines

Fig. 4(b) shows the effect of the number of machines. It tells that when the value of the machine number factor, μ , increases, ATCSSR outperforms ATCSR more. Additionally, for a given value of job machine factor, μ , using more machines yields more number of better cases.

6.3. Effect of the new part in the WSPT term of ATCSSR, $\max(s_{ij}, r_j - t)$

The result of the best of the best test summarized in Fig. 5(b) shows the importance of the new part in the WSPT term of ATCSSR. Without $\max(s_{ij}, r_j - t)$, the number of better cases is significantly reduced and the number of tied cases is increased.

7. Computation time

A larger search range and smaller grid size increase computation time while improving the quality of the final schedule; the computation time and quality of schedule need to be balanced in the application of the scheduling procedure. Table 4 shows the computation time of scheduling. All 5301 problems generated in Section 4.2 are tested on a 32 bit notebook with Pentium (1.86 GHz) processor and 1 GB RAM. We observe that using more scaling parameters improves the quality of results, but needs little bite more computation time. Table 4 also shows that the grid method is computationally fast enough for most real applications; however, a regression method will still be helpful when scheduling is needed soon.

8. Conclusions and future research

Depending on whether or not the setup is allowed to start before the ready time of a part, the sequence dependent setup is divided into CSDS and SSDS. This paper studies the performance of ATC-based rules on identical parallel machines under these two conditions. Results show that MATCSR and ATCSSR outperform other ATC-based rules, BATCS, BATCSmod, and ATCSR, in reducing the total weighted tardiness on identical parallel machines with unequal ready time when the sequence dependent setup is continuous and separable, respectively. The new parts in the WSPT terms of MATCSR and ATCSSR are shown to be significant. For the ATC-based indexes, using more scaling parameters improves the quality of results.

MATCSR and ATCSSR are also recommended to generate the initial solution and combined with other heuristics, such as tabu

search and simulated annealing, to find an improved solution, because the quality of the final solution is often highly dependent on the initial solution used (Tan et al., 2001).

Although the grid method is computationally fast enough for most real applications, a regression method will still be helpful when scheduling time is very binding; development of regression equations of the new formulas is needed. Development of an efficient heuristic procedure that considers continuous sequence dependent setup and separable sequence dependent setup simultaneously is also left for future research. Finally, under more complicated environments such as in a flow shop, in a job shop, or in a batch shop, performances of recently proposed ATC-based rules are still unknown. It is also necessary to evaluate their performances under these complex environments.

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